

JERRY CHU

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EDUCATION

University of California, Berkeley

B.A. Data Science, B.A Computer Science

Expected Graduation: May 2028

GPA: 3.93

- **Relevant Courses:** Computer Security, Efficient Algorithms and Intractable Problems, Data Structures, Principles & Techniques of Data Science, Computer Architecture/Machine Structures, Discrete Mathematics and Probability Theory

PROFESSIONAL EXPERIENCE

SLB

Sunnyvale, CA

Digital Technologies Intern - Full Stack Development

June 2026 - Present

- Built an agentic AI pipeline (OpenAI Agents SDK, GPT-5.4) with structured JSON output and multi-modal feedback loops to generate and refine slides, cutting slide creation from 2+ hours to under 3 minutes.
- Developed a Widget Factory (Pydantic v2, Jinja2, YAML) that mines agentic session traces into reusable interactive templates, reducing template dev from 5 days to a single CLI command (~\$0.12/widget).
- Shipped a full-stack platform (Streamlit, FastAPI/WebSocket, SQLite, Playwright) with browser-editable exports; adopted by 4 teams (40+ engineers), eliminating ~120 hours/quarter of manual PowerPoint work.

Ti Lab at Berkeley

Berkeley, CA

UC Berkeley EECS Undergraduate Researcher

January 2026 - Present

- Built a Python pipeline to automate DNA scaffold strand routing, engineering 15+ novel 3D geometries at 78% thermodynamic yield.
- Applied PyTorch/TensorFlow models to optimize one-pot reaction conditions, predicting non-interacting DNA bond sets to prevent misfolding in 10,000+ nucleotide structures at 94% accuracy.
- Built a React/Node.js Full-stack dashboard visualizing 3D nanostructure simulations for a team of 10+ researchers.

Anchor Logics

Berkeley, CA

Software Engineering Intern

May 2025 – August 2025

- Shipped a patient-facing Single-page Application (React/Redux, Node.js) for 300+ patients to securely access and visualize wearable medical data, cutting data retrieval requests 25%.
- Engineered a demo request page (TypeScript, Next.js) with form validation and Salesforce CRM integration to streamline sales pipeline.
- Operationalized the OpenPose ML model for real-time gait detection on AWS (S3) to optimize stimuli weight placement for the company's PIEZO2 medical wearable weighted vest.

PROJECTS

BerkeleyTime | *React, Typescript MongoDB, Docker, GraphQL*

- Contributed to open-source UC Berkeley course enrollment tool used by 20,000+ students/semester
- Built and maintained a full-stack staff dashboard used by 30+ staff members to manage logistics, refactoring legacy pages and APIs into modular, scalable components.
- Enhanced the beta Course Scheduler with historical grade distributions, enrollment trends, and four-year plan visualizations.

Encrypted File Storage Client | *Golang*

- Designed an end-to-end encrypted file sharing over an untrusted, active-adversary datastore
- Implemented Encrypt-then-MAC (AES-CTR + HMAC-SHA512) for confidentiality and integrity, achieving IND-CCA2 security.
- Designed capability-based access control (RSA-OAEP key exchange, RSA-PSS signatures) for multi-user sharing and secure revocation.

Cal Men's Tennis Analytics & SwingVision Prediction Model | *Python, Tensorflow, SwingVision, Pandas, Seaborn*

- Built a data pipeline from SwingVision video to structured CSV, then modeled predictive swing features (court heatmaps, ball/swing speed) at 88% classification accuracy.
- Translated feature importance into performance heuristics to optimize Cal Tennis players' shot selection.

TECHNICAL SKILLS

Skills: Java, Python, JavaScript, C/C++, HTML, CSS, SQL, Typescript, React, Node.js, GraphQL, MongoDB, Golang, x86, CAD

Tools: Claude, Cursor, Git, Visual Studio Code, Microsoft Office, Docker, AWS, Tableau, AutoDesk Fusion, Cura

Languages: English, Chinese, Japanese